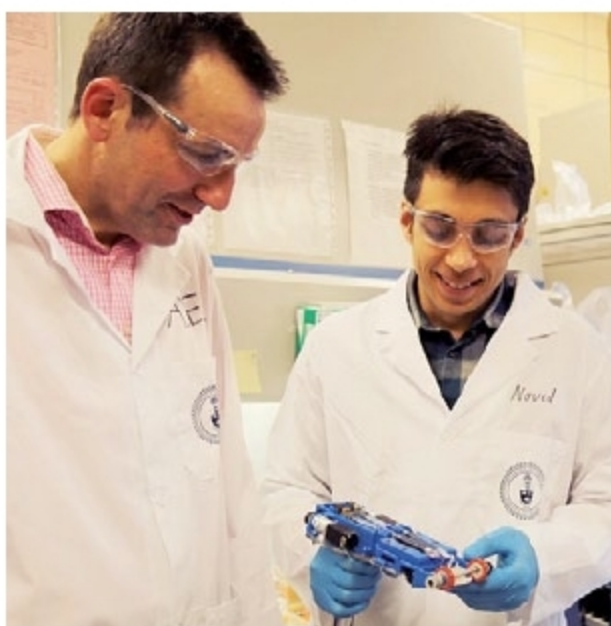


Portable 3D skin printer for repairing deep wounds

Researchers from University of Toronto, Canada, have developed a handheld 3D skin printer that deposits even layers of skin tissue to cover and heal deep wounds. The scientists believe it to be the first device that forms tissue *in situ*, depositing and setting in place, within two minutes or less.

The research was led by PhD student Navid Hakimi, under the supervision of associate professor Axel Guenther of Faculty of Applied Science & Engineering, and in collaboration with Dr Marc Jeschke, director of Ross Tilley Burn Centre at Sunnybrook Hospital and professor of immunology at Faculty of Medicine.

For patients with deep skin wounds, all three skin layers—epidermis, dermis and hypodermis—may be heavily damaged. The current preferred treatment is called split



Associate professor Axel Guenther and PhD student Navid Hakimi with the 3D skin printer
(Credit: University of Toronto)

thickness skin grafting, where healthy donor skin is grafted onto the surface epidermis and part of the underlying dermis. Split thickness grafting on large wounds requires enough healthy donor skin to traverse all three layers, but sufficient graft skin is rarely available. This leaves

a portion of the wounded area ungrafted, or uncovered, leading to poor healing outcomes.

The team believes that the *in situ* skin printer is a platform technology that can overcome these barriers, while improving the process of skin healing process. The printer resembles a white-out tape dispenser. Vertical stripes of bio ink—made up of protein-based biomaterials, including collagen, the most abundant protein in the dermis, and fibrin, a protein involved in wound healing—run along the inside of each tissue sheet.

Embroidered antennae, circuits for wearable electronics

Asimina Kiourti, assistant professor of electrical and computer engineering, Ohio State University, is working on making antennae and power sources that are flexible and washable. Specifically, the effort is to embroider electronics directly into fabrics using conductive threads, called e-threads.

The e-threads are bundles of twisted polymer filaments to provide strength, each with a metal-based coating to conduct electricity. The polymer core of each filament is made out of



Asimina Kiourti demonstrating her precision embroidery technique
(Credit: Ohio State University)

Kevlar or Zylon, while the surrounding coating is silver. Tens or even hundreds of these filaments are twisted together to form a single e-thread, which is usually less than half a millimetre across.

The e-threads can be easily used with common commercial embroidery equipment. The embroidered antennae are lightweight and just as good as their rigid copper counterparts, and can be as intricate as state-of-the-art printed circuit boards. e-thread antennae can even be combined with regular threads in more complex designs.

So far, smart hats have been created, which can read deep brain signals for patients with Parkinson's or epilepsy. Embroidered t-shirts with antennae extend the range of Wi-Fi signals to the wearers' mobilephones. Mats and bedsheets monitor infants' height to screen for a range of early childhood medical conditions. Foldable antennae measure how much of a surface the fabric is on has bent or lifted.

Ductile semiconductor for flexible electronics

Ductility is common in metals, while it is rarely observed in inorganic semiconductors, especially at room temperature. Most semiconductor materials are brittle and cannot bend, pull or compress.

Scientists from Shanghai Institute of Ceramics of Chinese Academy of Science, China, and Max-Planck-Institut für Chemische Physik fester Stoffe in Dresden, Germany, have found a type of inorganic semiconductor, with good ductility and flexibility at room temperature.

The newly-discovered semiconductor a-Ag₂S is ductile like metal and easily changes shape instead of breaking under outer deformation. It is expected to be applied in flexible electronic technologies such as wearable devices.

For applications in flexible electronics, the research team also produced a thin film made of a-Ag₂S and found that it had great deformability and good electrical properties.